

KRAFT

Knowledge-Reasoned Allocation For Teams

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Problem

Task allocation is a huge part of project management. There are currently many project management tools like Jira and Asana that help divide tasks and give visibility on team members' workloads. However, they still require human intervention and review to make the assignments.

Our Solution: KRAFT

KRAFT, or Knowledge-Reasoned Allocation For Teams, is an **AI-powered task allocation system**. It uses structured knowledge and FOPC-style rules to automatically assign work to team members based on their skills, availability, and workload. KRAFT also explains the rationale behind each allocation.

REPRESENTATION	REASONING	OUTPUT
FOPC-style rules over a structured knowledge base	Backward chaining inference + multi-criteria scoring	Task assignment with verifiable rationale

KRAFT Goes Beyond Current LLMs

KRAFT can handle the limitations that LLMs have when it comes to task allocation:

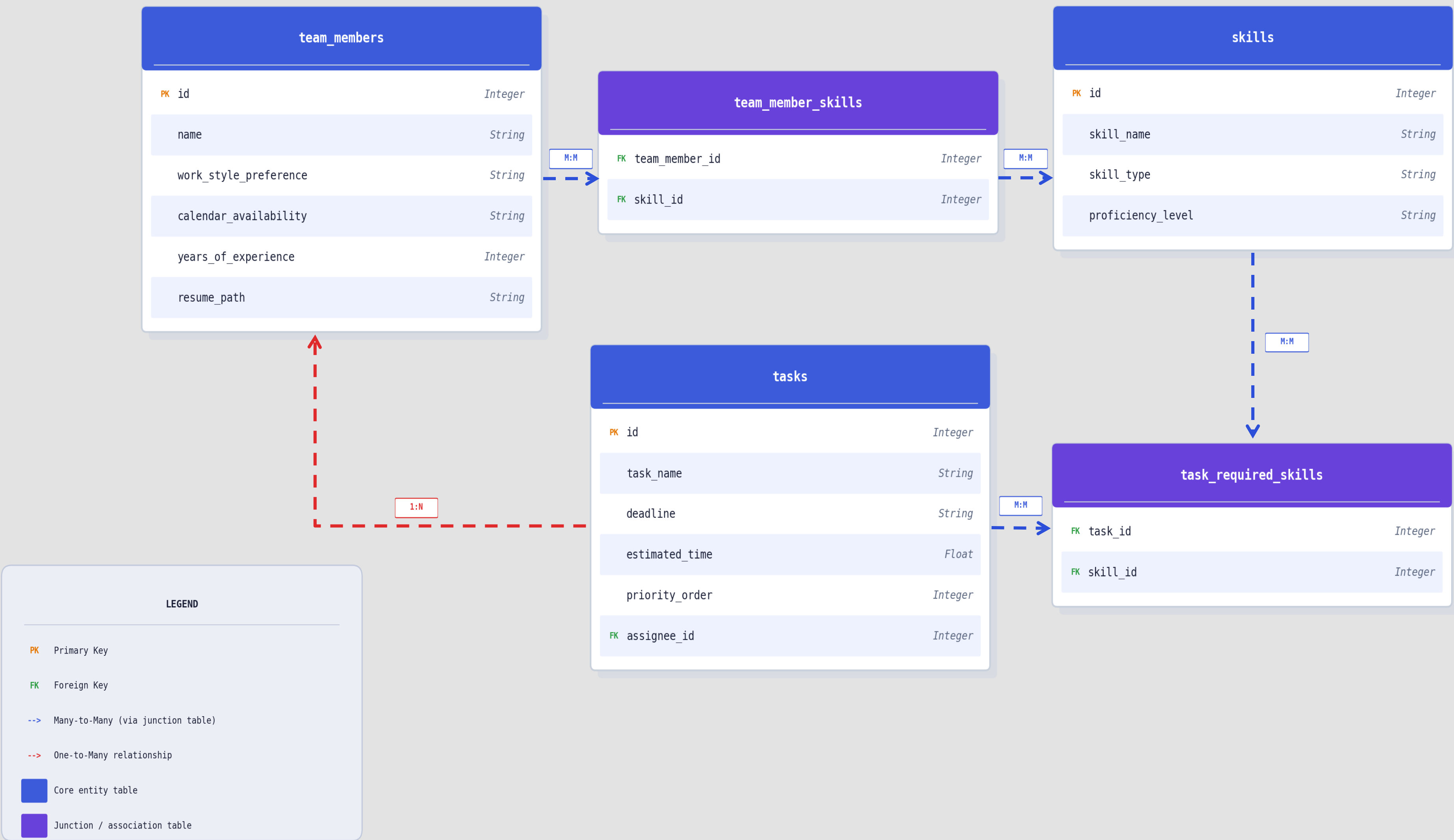
- **Maintains persistent, structured knowledge:** team members, skills, workloads and deadlines are represented explicitly, not held in context
- **Enforces hard constraints:** workload limits and required skills are guaranteed, not approximated
- **No hallucinations:** all assignments are derived from verifiable facts

Why KRAFT?

- **Task allocation is a high-stakes decision hiding in plain sight.**
Who gets the high-visibility project, who gets overloaded, who gets overlooked - these choices shape careers, burnout, and fairness. Yet most teams treat them as casual management calls, with no audit trail and no accountability.
- **LLMs explain after the fact. KRAFT explains by design.**
When an LLM says "Alice is the best fit," that's a plausible story - not the actual mechanism behind the decision. Ask again and the story changes. In KRAFT, the inference trace *is* the explanation: every assignment is derived from verifiable FOPC rules, constraint checks, and scoring algorithm, making "why X over Y" a provable chain, not a generated narrative.
- **The bottleneck for AI adoption isn't capability - it's trust.**
Organizations don't need smarter allocation; they need allocation they can defend, audit, and correct. KRAFT encodes fairness as checkable rules, so auditing means verifying policy compliance - not guessing whether a model "feels right."

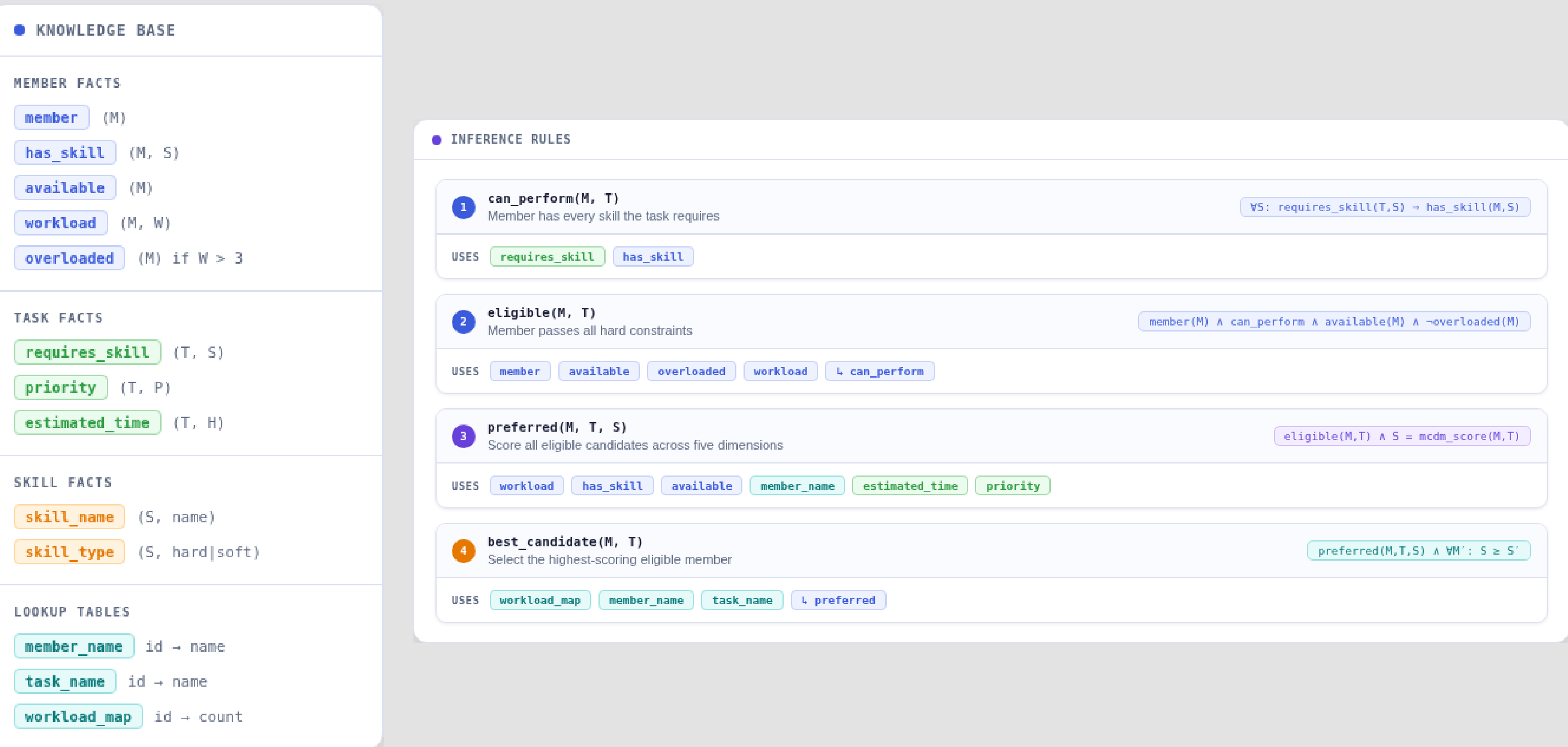
Database Schema

The relational data model captures team member profiles, skill sets, and task requirements to aid task allocation



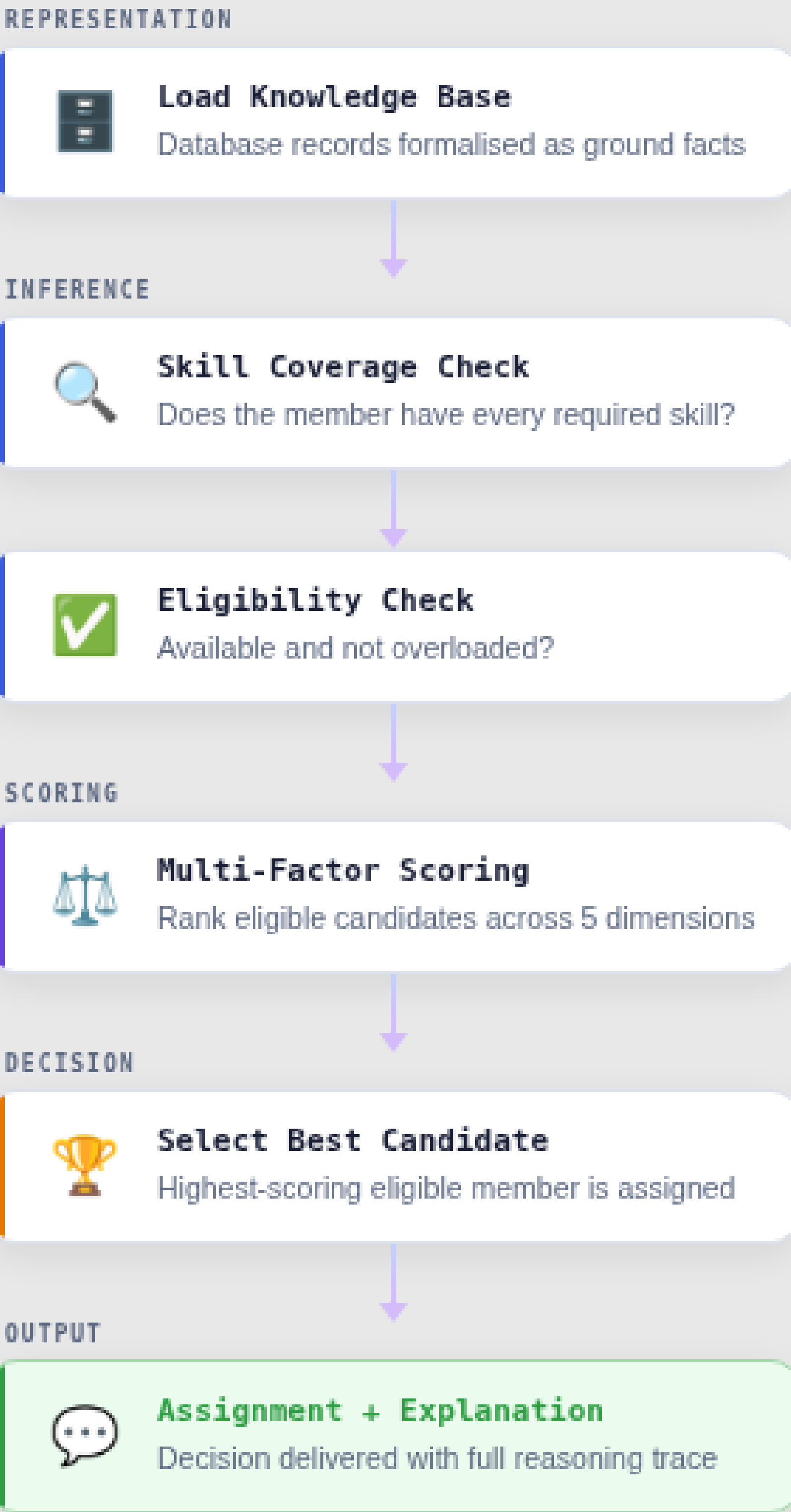
Knowledge Representation

Facts are loaded from the database and used to drive inference



Reasoning Pipeline

Logical inference through backward chaining to determine eligibility, followed by MCDM scoring to select the best candidate



Try it out!

TO DO